

Chain Rules [ใช้กับโจทย์ที่มองเป็นก้อนได้]

สูตร — (diff นอก)(diff ใน)

① $y = (2x+1)^2$

method $2(2x+1)^2 \cdot 2$
 $= (4x+2) \cdot 2$
 $= 8x+4$ #

② $\sqrt{x^3+7x+1}$

method $(x^3+7x+1)^{\frac{1}{2}}$
 $= \frac{1}{2}(x^3+7x+1)^{-\frac{1}{2}}(3x^2+7)$
 $= \frac{1}{2} \frac{3x^2+7}{\sqrt{x^3+7x+1}}$ #

Exponential [diff ขว้าง]

สูตร — $e^x = e^x$
 $e^u = e^u(u')$

① $\frac{d}{dx}[e^{-\frac{3}{x}}]$

method $\left[\frac{3}{x^2}\right]e^{-\frac{3}{x}} = \frac{3e^{-\frac{3}{x}}}{x^2}$

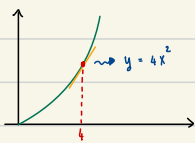
Note!

$\frac{-3}{x^2} \frac{d}{dx} = 3x^{-1} = 3x^{-2} = \frac{3}{x^2}$

Slope

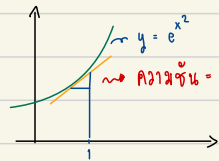
$m = \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$

①



$m = \frac{dy}{dx} = \frac{d(4x^2)}{dx}$
 $m = 8x$
 $\therefore m = 8(4) = 32$

②



$m = \frac{dy}{dx} = \frac{d(e^{x^2})}{dx}$
 $= e^{x^2} \cdot 2x$
 $m = e^{1^2} \cdot (2(1)) = 2e \approx 5.437$ #

Definite Integral [อินทิเกรตจำกัดเขต]

$\int_a^b f(x) dx$ — การหาขนาดของพื้นที่ $a \rightarrow b$

การหาค่า:

$\int f(x) dx = F(x) + c$

$\therefore \int_a^b f(x) dx = F(b) - F(a) \rightarrow$ ได้พื้นที่ใต้กราฟ

Note!

วิธีคือ 1. Integrate 2. แทนค่า 3. ลบกัน

① $\int_{-1}^2 3x^2 dx$

method $\int 3x^2 dx = \frac{3x^3}{3} + c$

Note!

How to Integrate: $\frac{x^{n+1}}{n+1} + c$ $\rightarrow$ constant ค่าคงที่

$\int_{-1}^2 3x^2 dx = [x^3]_{-1}^2$

① — แทน $x=2 = 2^3$

② — แทน $x=-1 = (-1)^3$

$0 - ① = 2^3 - (-1)^3$

$= 8+1 = 9$ #

② $\int_1^8 \sqrt[3]{x} dx$

method $\int x^{\frac{1}{3}} dx = \frac{x^{\frac{4}{3}}}{\frac{4}{3}} + c = \frac{3}{4}x^{\frac{4}{3}} + c$

$\int_1^8 \sqrt[3]{x} dx = \left[\frac{3}{4}x^{\frac{4}{3}}\right]_1^8$

① — แทน $x=8 = \frac{3}{4}(8)^{\frac{4}{3}}$

② — แทน $x=1 = \frac{3}{4}(1)^{\frac{4}{3}}$

$0 - ② = \frac{3}{4}(8)^{\frac{4}{3}} - \frac{3}{4}(1)^{\frac{4}{3}}$
 $= \frac{3}{4}(16) - 1$
 $= 12-1 = 11$ #

③ $\int_1^4 \sqrt{x} (1+2\sqrt{x}) dx$

method $\int \sqrt{x} (1+2\sqrt{x}) dx = \int \sqrt{x} + 2x dx$
 $= \int x^{\frac{1}{2}} + 2x dx$
 $= \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{2x^2}{2} + C = \frac{2}{3} x^{\frac{3}{2}} + x^2 + C$
 $\int_1^4 \sqrt{x} (1+2\sqrt{x}) dx = \left[\frac{2}{3} x^{\frac{3}{2}} + x^2 \right]_1^4$

① — when $x = 4 = \frac{2}{3} \cdot 4^{\frac{3}{2}} + 4^2$

② — when $x = 1 = \frac{2}{3} + 1$

① - ② = $\left(\frac{2}{3} \cdot 4^{\frac{3}{2}} + 4^2 \right) - \left(\frac{2}{3} + 1 \right)$
 $= \frac{64}{3} - \frac{5}{3}$
 $= \frac{59}{3} \#$

L'hospital Rule [การหาขีดด้วยโลปีตาล]

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \Rightarrow \text{interminate form}$

Note!

interminate form = $\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty, 1^\infty$

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \Rightarrow \text{diffuse / diff ล้าง}$

① $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = \frac{1^2-1}{1-1} = \frac{0}{0}$

method $\frac{2x}{1} = 2(1) = 2 \#$

② $\lim_{t \rightarrow \infty} \frac{\ln(3t)}{t^2} = \frac{\ln(3(\infty))}{\infty^2} = \frac{\infty}{\infty}$

method $\frac{1}{2t} \cdot \cancel{3} = \frac{1}{2t}$
 $\lim_{t \rightarrow \infty} = \frac{1}{2t^2} = \frac{1}{\infty} = 0 \#$

③ $\lim_{x \rightarrow -4} \frac{\sin(\pi x)}{x^2-16} = \frac{\sin(-4\pi)}{16-16} = \frac{0}{0}$

method $\frac{\cos(\pi x) \cdot \pi}{2x}$

$\lim_{x \rightarrow -4} \frac{\cos(\pi x) \cdot \pi}{2x}$

$= \frac{\pi \cos(-4\pi)}{2(-4)}$

$= \frac{\pi}{-8} \#$

Substitution technic [เทคนิคแทนตัวแปร]

① $\int \sin(3x+1) dx$

Note! ห้ามให้ Integrate ง่ายขึ้น

method $u = 3x+1 \rightarrow \frac{du}{dx} = 3 \rightarrow dx = \frac{du}{3}$

$\int \sin u \frac{du}{3} = \frac{1}{3} \int \sin u du = -\frac{1}{3} \cos u + C = -\frac{1}{3} \cos 3x+1 + C \#$

② $\int x(2x^2+3)^{10} dx$

method $u = 2x^2+3 \rightarrow \frac{du}{dx} = 4x \rightarrow dx = \frac{du}{4x}$

$\int \cancel{x} u^{10} \frac{du}{4x} = \frac{1}{4} \int u^{10} du = \frac{1}{4} \frac{u^{11}}{11} + C = \frac{1}{4} \frac{(2x^2+3)^{11}}{11} + C$

③ $\int 2x^2 \sqrt{x^3-4} \, dx$

method $u = x^3 - 4 \Rightarrow \frac{du}{dx} = 3x^2 \Rightarrow dx = \frac{du}{3x^2}$

$$\int 2x^2 (u)^{\frac{1}{2}} \frac{du}{3x^2} \rightarrow \frac{2}{3} \int u^{\frac{1}{2}} du = \frac{2}{3} \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{4}{9} u^{\frac{3}{2}} + C = \frac{4}{9} (x^3-4)^{\frac{3}{2}} + C \quad \#$$

Integrate Exponential

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

Note! $\int e^x dx = \frac{e^x}{\ln e} + C = e^x + C \quad \#$
 $\ln e = 1$

① $\int_1^3 2^x dx$

method $\int_1^3 2^x dx = \left[\frac{2^x}{\ln 2} \right]_1^3$

① — $\ln 2 \cdot x = 3 = \frac{2^3}{\ln 2}$
 ② — $\ln 2 \cdot x = 1 = \frac{2^1}{\ln 2}$

$$\textcircled{1} - \textcircled{2} = \frac{2^3}{\ln 2} - \frac{2^1}{\ln 2} = \frac{6}{\ln 2} \quad \#$$

② $\int x \cdot 3^{x^2+1} dx$

method $u = x^2 + 1 \rightarrow \frac{du}{dx} = 2x \rightarrow dx = \frac{du}{2x}$

$$\int x \cdot 3^u \frac{du}{2x} = \frac{1}{2} \int 3^u du = \frac{1}{2} \frac{3^u}{\ln 3} = \frac{1}{2} \frac{3^{x^2+1}}{\ln 3} + C \quad \#$$

③ $\int x^2 e^{x^3+1} dx$

method $u = x^3 + 1 \rightarrow \frac{du}{dx} = 3x^2 \rightarrow dx = \frac{du}{3x^2}$

$$\int x^2 e^u \frac{du}{3x^2} = \frac{1}{3} \int e^u du = \frac{1}{3} \frac{e^u}{\ln e} = \frac{1}{3} e^{x^3+1} + C \quad \#$$

diff trigonometry

$\sin / \cos / \tan \quad | \quad \csc / \sec / \cot$
 $\frac{d}{dx} \sin x = \cos x \quad | \quad \frac{d}{dx} \cos x = -\sin x$

① $\frac{d}{dx} \sin(x^2 - 2x)$

method $\cos(x^2 - 2x) \cdot (2x - 2) \quad \# \quad \# \text{ Chain Rule!}$

② $\frac{d}{dx} (\sin(x^2 - 2x + 1) \cdot \cos^2 x)$
 $u = \cos^2 x \rightarrow -2 \cdot \cos x \cdot \sin x \quad \# \text{ Chain Rule}$

method $[\sin(x^2 - 2x + 1) \cdot (-2 \cdot \cos x \cdot \sin x)] + [\cos^2 x \cdot (\cos(x^2 - 2x + 1) \cdot (2x - 2))] \quad \# \text{ Product Rule! } [u \cdot v]' = [u]' \cdot v + u \cdot [v]'$

indeterminate form $\left[\frac{0}{0}, \frac{\infty}{\infty}, 0 \cdot \infty, \infty - \infty \right]$

- 1) แยกตัวประกอบ
- 2) Conjugate
3. L' hospital

1) แยกตัวประกอบ

$$\textcircled{1} \lim_{x \rightarrow 3} \frac{x^2 + x - 12}{3 - x} = \frac{3^2 + 3 - 12}{3 - 3} = \frac{0}{0}$$

method $\frac{(x+4)(x-3)}{3-x} = \lim_{x \rightarrow 3} -x+4 = -3+4 = -1 \neq$

2) Conjugate # ตัวคูณที่ทำให้ส่วนตรง

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{x}{2 - \sqrt{x+4}} = \frac{0}{2-2} = \frac{0}{0}$$

method $\frac{x}{2 - \sqrt{x+4}} \cdot \frac{2 + \sqrt{x+4}}{2 + \sqrt{x+4}} = \lim_{x \rightarrow 0} \frac{x(2 + \sqrt{x+4})}{2^2 - (\sqrt{x+4})^2} = \lim_{x \rightarrow 0} \frac{x(2 + \sqrt{x+4})}{4 - (x+4)} = \lim_{x \rightarrow 0} \frac{x(2 + \sqrt{x+4})}{-x} = \lim_{x \rightarrow 0} -(2 + \sqrt{x+4}) = -2 - \sqrt{0+4} = -2 - 2 = -4 \neq$

find limit with วิธีแทนค่า

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{\sin 0}{0} = \frac{0}{0} \quad \# \text{ สมมติ } x \text{ จะเข้าด้วยค่าที่เข้าใกล้ 0 มากๆ } * \text{ need calculator } *$$

method สมมติ $x = 10^{-9} = \frac{\sin 10^{-9}}{10^{-9}} = 1 \neq$

$$\textcircled{2} \lim_{x \rightarrow 3} \frac{1 - \cos(x-3)}{x^2 - x - 6} = \frac{0}{0}$$

method แทน $x \rightarrow 3$ มากๆ $= 3 + 10^{-9} = 0 \neq \quad \# \text{ use calculator only!}$

Partial derivative [อนุพันธ์ย่อย]

↳ derivative ของ function หลายตัวแปร

$$y = f(x_1, x_2, x_3)$$

$$\frac{\partial y}{\partial x_1}, \frac{\partial y}{\partial x_2}, \frac{\partial y}{\partial x_3} \Rightarrow \text{การเปลี่ยนแปลงของ } y \text{ เทียบกับ } x \text{ แต่ละตัว} = \text{partial derivative}$$

Note! ถ้าเทียบการเปลี่ยนแปลงของ y กับ $x_1 \rightarrow x_2, x_3$ ไม่เปลี่ยน (เป็นค่าคงที่)

$$\textcircled{1} f = x_1^2 + 2x_1x_2 - x_2^2$$

method $\frac{\partial f}{\partial x_1} = \frac{\partial}{\partial x_1} (x_1^2 + 2x_1x_2 - x_2^2)$

$$= 2x_1 + 2x_2(1) - 0$$

$$= 2x_1 + 2x_2 \neq$$

$$\textcircled{2} f(x, y) = 3x^5y^{10} + 2y^3 - x^4 \text{ find } f_x(x, y), f_y(x, y)$$

method $\frac{\partial}{\partial x} [3x^5y^{10} + 2y^3 - x^4] = 15x^4y^{10} + 0 - 4x^3 \neq$

$$\frac{\partial}{\partial y} [3x^5y^{10} + 2y^3 - x^4] = 3x^5 \cdot 10y^9 + 6y^2 - 0 \neq$$

Extrema

1) หา Critical Number

$$\rightarrow f'(x)$$

$$\rightarrow \text{หา } f'(x) \text{ หา } = 0$$

2) แทน Critical Number ใน $f(x)$

3) แทนค่าซ้าย-ขวาใน $f(x)$

4) เปรียบเทียบค่าที่แทนทั้งหมดเพื่อหาค่า MAX, MIN

Note!

$[m, n] \rightarrow$ ช่วงปิด, Continue = หา MAX, MIN ได้

$(m, n) \rightarrow$ ช่วงเปิด, Continue = หา MAX ไม่ได้

$[m, n] \rightarrow$ ช่วงปิด, ไม่ Continue = หา MIN ไม่ได้

① $f(x) = 3x^4 - 4x^3$ on the interval $[-1, 2]$

method $f'(x) = 12x^3 - 12x^2$
 $f'(x) = 12x^3 - 12x^2 = 0$ # $12x^2 = 0$
 $= 12x^2(x-1) = 0$ # $12x^2$ factor
 $x = 0, 1$ # Critical Number

$f(-1)$ $\leftarrow$ $f(-1)$ $\leftarrow$ Left = 7
 $f(2)$ $\leftarrow$ Right = 16 # MAX
 $f(0)$ c-number = 0
 $f(1)$ c-number = -1 # MIN

} $\leftarrow$ $\leftarrow$ MIN / MAX

Continuity

① $f(x) = \begin{cases} 2x+1, & x < 2 \\ 5, & x = 2 \\ 3x-1, & x > 2 \end{cases}$ $f(x)$ Continue on $(0, 4)$?

method $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} [2x+1] : \text{when } x=2 = 2(2)+1 = 5$

$\lim_{x \rightarrow 2^+} [3x-1] : \text{when } x=2 = 3(2)-1 = 5$

$f(x)$ $\leftarrow$ $x=2 = 5$

$\therefore$ $\leftarrow$ $\leftarrow$ Continuity #

Note!

$f(x)$ continue on $[a, b]$

1) continue on (a, b) ?

2) $f(a) = \lim_{x \rightarrow a^+} f(x)$

$f(x)$ continue on $(a, b]$

1) continue on (a, b) ?

2) $f(b) = \lim_{x \rightarrow b^-} f(x)$

② $f(x) = \begin{cases} x^3 - 4x & ; x \leq -1 \\ 2x + 5 & ; x > -1 \end{cases}$ Continue $[-1, 3)$?

method 1) $(-1, 3)$ Continue

2) $f(-1) = \lim_{x \rightarrow -1^+} f(x)$

$\therefore f(-1) = (-1)^3 - 4(-1) = -1 + 4 = 3$

$\lim_{x \rightarrow -1^+} (2x+5) = 2(-1)+5 = 3$

$\therefore$ $\leftarrow$ $\leftarrow$ Continue on $[-1, 3)$

Implicit diff

$\leftarrow$ $\leftarrow$ $\frac{dy}{dx}$ $\leftarrow$ $f(x, y) = 0$

① $x^2 + y^2 = 2$ $\frac{dy}{dx} = ?$

method $\frac{d}{dx} (x^2 + y^2) = \frac{d}{dx} (2)$

$2x + 2y \cdot \frac{dy}{dx} = 0$

$\frac{dy}{dx} = \frac{-2x}{2y} = \frac{-x}{y}$ #

$$\textcircled{2} \quad 2y^3 + 4x^2 - y = x^6 \quad \text{or} \quad \frac{dy}{dx}$$

method $\frac{d}{dx}(2y^3 + 4x^2 - y) = \frac{d}{dx}(x^6)$

$$6y^2 \frac{dy}{dx} + 8x - \frac{dy}{dx} = 6x^5$$

$$\frac{dy}{dx}[6y^2 - 1] = 6x^5 - 8x$$

$$\frac{dy}{dx} = \frac{6x^5 - 8x}{6y^2 - 1} \quad \#$$